

Z-Score Normalisation & Trading Signal Generation

zscore.py — Mathematical Notes & Documentation

Rolling / Expanding Normalisation, Threshold-Based State Machine, Signal Encoding

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Abstract. The raw spread S_t has units of log-price difference and its scale varies across pairs and over time. The Z-score $Z_t = (S_t - \mu_t)/\sigma_t$ converts S_t to a universal, dimensionless scale with interpretable thresholds. This document derives the rolling and expanding Z-score formulas, explains the state-machine signal generator with entry, exit, and stop-loss rules, and shows how Z_t feeds into the Kalman filter as a normalised observation.

Why Z-Score?

A spread of $S_t = 0.05$ is meaningless without context. Is it large or small? The Z-score answers this:

Key Concept

$Z_t = 2.0$ means the spread is 2 standard deviations above its rolling mean. Under a normal distribution, values $|Z_t| > 2$ occur only $\sim 5\%$ of the time — indicating a significant deviation worth trading.

Z-Score Computation

Rolling Z-Score

Formula 1 — Rolling Z-Score

$$\mu_t^{(W)} = \frac{1}{W} \sum_{i=0}^{W-1} S_{t-i} \quad (1)$$

$$\sigma_t^{(W)} = \sqrt{\frac{1}{W-1} \sum_{i=0}^{W-1} (S_{t-i} - \mu_t^{(W)})^2} \quad (2)$$

$$Z_t = \frac{S_t - \mu_t^{(W)}}{\sigma_t^{(W)}} \quad (3)$$

W = lookback window $\approx 2\tau$ (from `half_life.py`).

Why $W \approx 2\tau$? The rolling window captures ~ 2 complete mean-reversion cycles, giving μ_t and σ_t enough data to be stable while remaining adaptive to regime changes.

Expanding Z-Score

Formula 2 — Expanding Z-Score

$$\mu_t^{(\infty)} = \frac{1}{t} \sum_{s=1}^t S_s \quad (4)$$

$$\sigma_t^{(\infty)} = \sqrt{\frac{1}{t-1} \sum_{s=1}^t (S_s - \mu_t^{(\infty)})^2} \quad (5)$$

$$Z_t = \frac{S_t - \mu_t^{(\infty)}}{\sigma_t^{(\infty)}} \quad (6)$$

Uses all history from $t = 1$. More stable but less adaptive.

Use for very slow mean-reverting pairs ($\tau > 120$ obs).

Comparison: Rolling vs. Expanding

Property	Rolling	Expanding
Adapts to regime	Yes	Slowly
Requires warm-up	W obs	$\min_periods$
Suitable τ	< 60 obs	> 120 obs
Signal stationarity	Better	Risk of drift

Signal Thresholds

Formula 3 — Threshold Zones

Zone	Condition	Action
LONG spread	$Z_t < -z_{\text{entry}}$	Buy Y, Sell X
SHORT spread	$Z_t > z_{\text{entry}}$	Sell Y, Buy X
EXIT / FLAT	$ Z_t < z_{\text{exit}}$	Close all
STOP-LOSS	$ Z_t > z_{\text{stop}}$	Force exit

Default values: $z_{\text{entry}} = 2.0$, $z_{\text{exit}} = 0.5$, $z_{\text{stop}} = 3.5$.

Why $z_{\text{entry}} = 2.0$? Under normality, $|Z_t| > 2$ occurs $\approx 5\%$ of the time — a statistically significant deviation. Tighter thresholds ($z_{\text{entry}} < 1.5$) trade more often but with lower edge per trade.

Why $z_{\text{exit}} = 0.5$? We close near the mean ($\mu \pm 0.5\sigma$) to capture the bulk of the reversion without waiting for a full return to zero (which may never happen exactly).

State Machine Signal Generator

Why a State Machine?

A naive signal would trigger repeatedly while $|Z_t| > 2$. The state machine prevents this: once a position is opened, it stays open until the exit condition is met, regardless of Z_t continuing to grow.

State Transitions

Formula 4 — Signal State Machine

States: $\Pi \in \{-1, 0, +1\}$, Override: Stop = ± 2 .

Each timestep t :

1. If $|Z_t| > z_{\text{stop}}$: emit $\text{sign}(Z_t) \times 2$; $\Pi \leftarrow 0$.
2. Elif $\Pi \neq 0$ and $|Z_t| < z_{\text{exit}}$: $\Pi \leftarrow 0$.
3. Elif $\Pi = 0$ and $Z_t < -z_{\text{entry}}$: $\Pi \leftarrow +1$.
4. Elif $\Pi = 0$ and $Z_t > z_{\text{entry}}$: $\Pi \leftarrow -1$.
5. Emit Π_t .

Signal encoding:

Signal	Name	Meaning
+1	LONG	Buy Y, Short X
-1	SHORT	Sell Y, Buy X
0	FLAT	No position
+2	STOP	Forced exit (Z extreme high)
-2	STOP	Forced exit (Z extreme low)

Code Flowchart

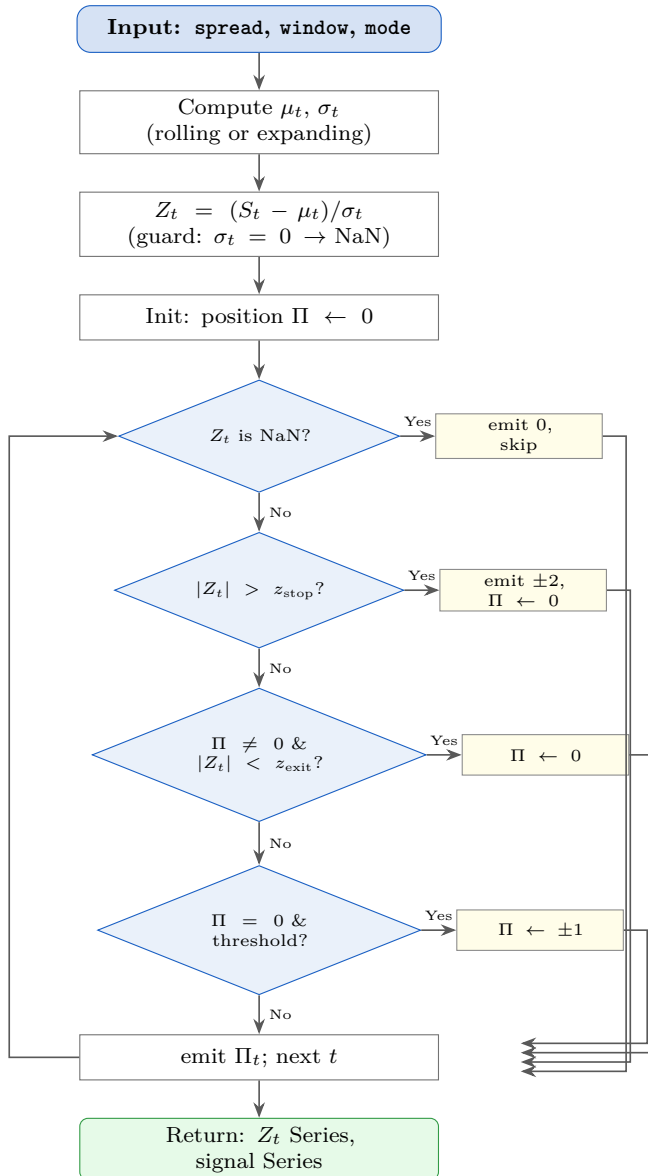


Figure 1: State machine for `ZScore.generate_signals()`. The loop-back arrow advances t at each iteration.

Worked Example

Setup

$S = [0.05, -0.10, 0.08, -0.06, 0.12, -0.03, 0.07]$
 Mode: rolling, $W = 3$, $z_e = 1.0$, $z_x = 0.3$, $z_s = 2.0$.

t	S_t	μ_t	σ_t	Z_t
3	0.08	0.010	0.076	0.92
4	-0.06	0.007	0.092	-0.73
5	0.12	0.047	0.093	0.79
6	-0.03	0.010	0.076	-0.52
7	0.07	0.053	0.076	0.23

With $z_e = 1.0$: no entry triggers (all $|Z_t| < 1.0$).

With $z_e = 0.7$: $Z_3 = 0.92 > 0.7$ triggers SHORT at $t = 3$.

Interpretation

No signal fired at $z_e = 2.0$ for this 7-point sample — the spread is mild. This is expected: with a half-life of ~ 1 observation, the spread oscillates rapidly but with small amplitude, giving Z_t values below the entry threshold.

Python Implementation

Z-score computation

```

def compute(self) -> pd.Series:
    s = self._spread
    if self.mode == "rolling":
        roller = s.rolling(window=self.window,
                           min_periods=self.min_periods)
        self._mu = roller.mean()
        self._sigma = roller.std(ddof=1)
    else: # expanding
        exp = s.expanding(min_periods=self.min_periods)
        self._mu = exp.mean()
        self._sigma = exp.std(ddof=1)
    sigma_safe = self._sigma.replace(0.0, np.nan)
    self._zscore = ((s - self._mu) / sigma_safe)
                    .rename("zscore")
    return self._zscore

```

State-machine signal (inner loop)

```

for t in range(n):
    zt = z[t]
    if np.isnan(zt): sig[t]=0; continue
    # stop loss
    if abs(zt) > self.stop_z:
        sig[t] = 2*int(np.sign(zt))
        position = 0; continue
    # exit
    if position != 0 and abs(zt) < self.exit_z:
        position = 0
    # entry
    if position == 0:
        if zt < -self.entry_z: position = 1
        elif zt > self.entry_z: position = -1
    sig[t] = position

```

Connection to Kalman Filter

ZScore $\rightarrow$ Kalman Interface

The Kalman Filter in `complete_kalman_filter.py` uses the *raw* spread S_t (not Z_t) as its observation. However, the Z-score is useful as a *post-filter* normalisation:

1. Kalman produces dynamic spread $s_t = Y_t - \hat{\beta}_t X_t$.
2. `ZScore(s_kalman, window=2*tau)` normalises it.
3. Z_t from Kalman is more stable as β_t adapts.

Recommended: use Kalman spread with rolling Z-score for live signals.

Threshold Selection Guide

Parameter	Default	Guideline
z_{entry}	2.0	Lower=more trades, less edge
z_{exit}	0.5	Higher=exit earlier, less profit
z_{stop}	3.5	Must $> z_{\text{entry}}$; 3σ is standard
W (window)	2τ	From <code>half_life.estimate()</code>
mode	rolling	Use expanding for $\tau > 120$